CS 498 Proposal: Calibration-Sufficiency-Aware Budget Allocation for MoE Post-Training Quantization

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1 What is the problem?

Post-training quantization (PTQ) reduces model memory requirements by converting high-precision weights to lower-bit representations without retraining. Methods such as GPTQ use calibration activations to estimate second-order information for weight quantization [1]. In Mixture-of-Experts (MoE) models, sparse routing causes different experts to receive unequal amounts of calibration data.

However, routed-token count may not fully describe calibration sufficiency. Experts with similar exposure can have different activation distributions and rates of statistical convergence. We therefore ask: **Can an expert's current calibration state predict the benefit of additional calibration, and can this information improve quantization quality under a fixed calibration budget?**

We define sufficiency through the marginal improvement obtained from more calibration data. Our goal is to direct a limited budget toward experts that still benefit from additional evidence, while retaining the standard GPTQ solver.

2 Why does it matter?

Calibration data can substantially affect PTQ quality [2]. This is particularly challenging for MoE models: sparse routing gives some experts many calibration tokens and leaves others with limited evidence. Poorly characterized experts can introduce quantization error when they are activated, even if the calibration set appears representative at the model level.

However, expert exposure is only part of the problem. Receiving more tokens does not necessarily mean that an expert's activation or second-order statistics are sufficiently reliable. Experts with similar routing counts may differ in activation variability and in how quickly additional data stops improving reconstruction. A uniform increase in the global budget can therefore collect redundant evidence for some experts while others still need more useful samples. Balancing token counts alone may also miss these differences.

Our project investigates whether this heterogeneity can guide calibration allocation. If current statistical state predicts the benefit of additional data, the same sequence budget could be used more effectively by prioritizing samples that activate experts with remaining calibration needs. The intended benefit is better quantization quality, especially for poorly calibrated experts, without changing the quantization solver. This also tests whether coverage is an adequate proxy for sufficiency, rather than assuming that rare experts always need more data or that frequently activated experts are already well calibrated.

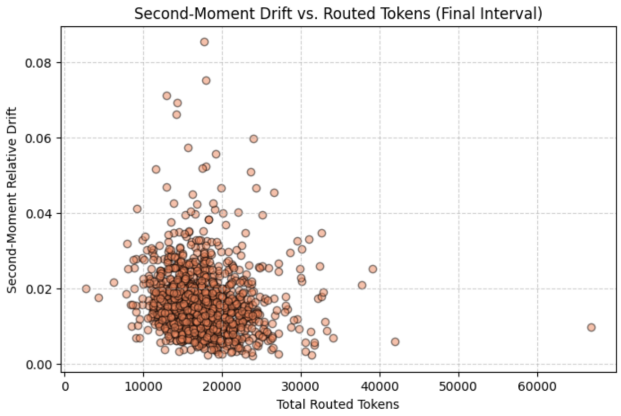


Figure 1: Routed tokens versus cumulative second-moment drift from 256 to 512 calibration sequences. Similar routing volume can correspond to substantially different statistical stability. Drift alone does not establish calibration sufficiency.

Preliminary observations. Our initial Qwen1.5-MoE-A2.7B experiments show that experts with similar routed-token counts can exhibit substantially different statistical drift (Figure 1). Pilot 4-bit GPTQ results also suggest that drift may predict subsequent reconstruction improvement at low budgets, especially from 64 to 128 sequences, while this relationship weakens at larger budgets. These findings require validation across more experts and seeds. Timing measurements indicate that GPTQ solving dominates the measured workload, limiting the runtime benefit of reducing calibration sequences. We therefore prioritize quantization quality at a fixed budget rather than runtime reduction.

Related work. MoEQuant addresses inter-expert imbalance through Expert-Balanced Self-Sampling and intra-expert imbalance through Affinity-Guided Quantization [3]. EAQuant combines expert-aware smoothing, routing consistency alignment, and expert-level calibration data balance [4]. Our complementary focus is the remaining value of calibration: we aim to allocate data according to

an expert's evolving statistical state rather than coverage alone. We retain standard GPTQ, while testing whether this state provides predictive information beyond routing exposure.

3 How do you intend to solve it?

We plan to build a **calibration-sufficiency-aware PTQ pipeline** with three components. From a larger candidate pool $\mathcal{D}$, the pipeline selects a calibration subset $\mathcal{C} \subseteq \mathcal{D}$ with $|\mathcal{C}| = B$, where B is the fixed sequence budget.

Routing and statistics profiler. The profiler records routing information for candidate sequences and incrementally tracks routed-token counts, activation moments, and Hessian summaries for the selected data. Statistics use activations actually routed to each expert. Drift compares token-count-normalized moments, such as $\bar{H}_e = H_e/n_e$, so changes do not simply reflect growth in the cumulative token count. Compact summaries support a single-GPU implementation.

Calibration sufficiency estimator. The estimator outputs a remaining-need score u_e for each expert: larger values indicate greater expected benefit from additional calibration. Candidate signals include activation drift and Hessian summaries. Offline characterization across experts and budgets will use held-out GPTQ reconstruction improvement to identify signals that predict future benefit beyond routing count. The pipeline will use a validated, low-overhead score as a proxy for this benefit; statistical stability alone will not be assumed to establish sufficiency.

Adaptive calibration controller. Starting from an initial calibration subset, the controller combines the scores u_e with candidate routing profiles to prioritize unselected sequences that activate experts with greater remaining needs. After each selected batch, the profiler updates expert statistics and the estimator refreshes its scores. This feedback loop continues until $|\mathcal{C}| = B$, including the initial subset. Tokens retain the model's natural routing. Global early stopping remains an optional extension.

Finally, standard GPTQ quantizes the model using the accumulated calibration statistics. The main intervention is how calibration evidence is acquired before quantization. Profiling and selection overhead will be included in evaluation.

4 How will you evaluate your solution?

Our primary setting is Qwen1.5-MoE-A2.7B with 4-bit weight-only GPTQ on a single GPU. Calibration data will come from standard corpora such as WikiText-2 or C4. We will use separate calibration, tuning, and evaluation data; an additional model or corpus is a stretch goal.

Characterization. We will compare how well routing counts and statistical signals predict future reconstruction improvement across budgets. Correlations, expert error curves, and count-matched comparisons will assess whether statistical state provides information beyond exposure. Signals will use only data available at the starting budget. Reconstruction will be evaluated on fixed held-out activations, with original weights restored before each independent quantization run.

Fixed-budget comparison. We will compare three calibration strategies:

- **Random:** conventional random sample selection.
- **Coverage-aware:** prioritize samples using expert routing counts.
- **Sufficiency-aware:** prioritize samples using validated expert calibration-state signals.

The coverage baseline isolates the value of the sufficiency signal; it is not a full reproduction of MoEQuant or EAQuant.

Our primary metrics are held-out expert reconstruction error, including average and tail errors, and model-level perplexity. We will first test a low-budget setting, then expand to budgets such as 64, 128, 256, and 512 sequences. All strategies will share the calibration pool, sequence length, quantization settings, and hardware. Multiple seeds and signal ablations will assess robustness. GPU time, peak memory, profiling, and selection overhead will be secondary metrics: equal selected-data budgets do not necessarily imply equal total compute.

5 What is the timeline?

1. **Oct. 1–15:** Complete expert-level characterization and identify promising sufficiency signals.
2. **Oct. 16–Nov. 4:** Implement fixed-budget allocation and random/coverage-aware baselines.
3. **Nov. 5–25:** Run reconstruction, perplexity, ablation, and overhead experiments.
4. **Nov. 26–Dec. 18:** Complete analysis, report, and presentation; add extensions if time permits.

References

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